

U.S. Tax Reform and International Trade

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July 2026

Abstract

We examine how changes in the taxation of multinational income affect U.S. firms' cross-border trade in physical goods. We study this question in the context of the 2017 U.S. tax reform, which substantially changed the tax treatment of multinational income. To examine whether these changes translated into real trade responses, we use firm-level shipping container data that capture physical goods exports and imports. Using a quasi-difference-in-differences approach, we find that after the tax reform, U.S. firms that are more exposed to the reform's international tax provisions significantly increase their physical goods exports and imports relative to less-exposed firms. The effects are stronger for firms with higher pre-tax profitability, greater R&D intensity, and lower shipping costs, consistent with a tax-based explanation. Furthermore, we document an increase in intersegment sales across geographic segments. Our study provides the first empirical evidence on the tax reform's effect on U.S. firms' cross-border trade in physical goods.

Keywords: corporate taxes, international trade, tax reform

JEL Classification: F6, F14, F23, H25, H26

We thank Matti Boie-Wegener (discussant), Marco Errico (discussant), John Gallellmore, David Godsell, Jeffrey Hoopes, Trent Krupa, Morris Liu, Nanqin Liu, Jeri Seidman, Gang Wang, Juanting Wang, Liangliang Wang, Anthony Welsch, Yuchen Wu, Jason Xiao, Jing Xing, Wenjian Xu, Will Yoder, participants at the 12th Annual Mannheim Taxation Conference, AAA International Accounting Section Research Webinar Series, the 2nd Mediterranean Accounting Conference, and seminar participants from the University of Macau, Nanjing University, Shanghai Jiao Tong University, and the Shanghai University of Finance and Economics for their valuable comments and suggestions. Travis Chow, Guoman She, and Xuanpu Lin acknowledge financial support from the University of Hong Kong. Guoman She acknowledges financial support from the National University of Singapore. Edward Maydew acknowledges the financial support from the University of North Carolina at Chapel Hill.

1. Introduction

We examine how changes in the taxation of multinational income affect U.S. firms' cross-border trade in physical goods. We study this question in the context of the Tax Cuts and Jobs Act (TCJA), enacted in 2017, which fundamentally reformed the U.S. taxation of multinational income. Policymakers and legislative discussions often framed the law's international provisions as intended to improve the global competitiveness of U.S. firms and to strengthen the United States as a location from which U.S. firms compete in foreign markets (Polantz 2020; Cole 2024; Urban-Brookings Tax Policy Center 2024a). Related policy commentary emphasized that the competitiveness effects of corporate tax policy are especially important for innovation-based traded sectors that compete in global markets (Atkinson 2021). Whether these provisions translated into measurable changes in firms' real cross-border trade, however, remains an open empirical question.¹

We focus on physical goods trade flows because they are an important real channel through which firms operate internationally. Trade affects firms' access to foreign demand, input costs, input availability, and supply chain organization, and can therefore affect firm growth and domestic investment (De Loecker 2007; Halpern, Koren, and Szeidl 2015; Pierce and Schott 2016). This focus is also policy relevant: U.S. government agencies routinely frame trade as central to U.S. economic performance and competitiveness (U.S. Department of Commerce 2024; Office of the United States Trade Representative 2024). International trade is also a major component of the U.S. economy: In 2025, U.S. exports and imports of goods

¹ The 'Unified Framework for Fixing Our Broken Tax Code' released by the U.S. Treasury in 2017 emphasizes reducing distortions created by the prior worldwide tax system and mitigating incentives for profit shifting and offshore investment (U.S. Treasury 2017). Congressional discussions typically frame the TCJA's international provisions as an effort to enhance U.S. firms' ability to compete and win in global markets. For example, the U.S. Senate, Committee on Finance highlighted that manufacturers had been "given license to compete more fiercely in the global economy" (U.S. Senate, Committee on Finance 2018), and Senator Crapo described "the goal of our new system" as "to both ensure the U.S., and U.S. companies, are competitive in the global marketplace and to protect the U.S. tax base" (U.S. Senate, Committee on Finance 2021, 69–70).

and services totalled \$7.76 trillion, equal to about 24.7% of U.S. GDP.² Although this aggregate includes both goods and services, our empirical analysis focuses specifically on cross-border trade in physical goods. This setting is economically important: S&P Global Market Intelligence reports that liner shipping transported more than \$1.5 trillion in U.S. trade in 2023, representing about 30% of total U.S. trade across transportation modes, and supported 9 million U.S. jobs (Bingham, Flanagan, and McLaughlin 2025). Physical goods trade is therefore a useful setting for evaluating one observable real response through which the tax reform may have affected firm behavior.

Prior research examines the reform's effects on corporate tax burdens and investment. To our knowledge, however, no study provides direct evidence on whether the reform affects firms' cross-border physical trade.³ We address this gap in the literature by examining how the tax reform affects U.S. firms' physical goods exports and imports using detailed shipment-level container data from the S&P Global Panjiva database (Smirnyagin and Tsyvinski 2022; Bisetti, She, and Zaldokas 2026; Chow, Maydew, and She 2026). These data allow us to observe cross-border shipments between firms and their foreign counterparties, including shipment origins and destinations, product codes, and product descriptions. Because the data capture the physical movement of goods across borders, our analysis speaks directly to real responses rather than to changes in how firms report or classify foreign activity (Desai and Hines 2001).

The TCJA tax reform introduces several major changes to the U.S. international tax regime that could affect how U.S. firms organize cross-border activity. First, the reduction in the federal corporate tax rate from 35% to 21% narrowed the tax wedge between domestic and foreign income, reducing incentives to allocate income to low-tax jurisdictions. Second, the deduction for Foreign-Derived Intangible Income (FDII) lowers the effective tax rate on certain

² U.S. Bureau of Economic Analysis (2025; 2026); Total trade is calculated as exports (\$3.43 trillion) plus imports (\$4.33 trillion); the trade share of GDP is calculated using annual current-dollar GDP (about \$31.42 trillion).

³ We review the relevant literature on the effect of the TCJA in Section 2.1.

foreign-derived income earned by U.S. corporations, increasing the attractiveness of serving foreign customers from the United States. Third, the Global Intangible Low-Taxed Income (GILTI) regime imposes a minimum tax on certain low-taxed foreign income associated with mobile intangible assets, weakening incentives to earn mobile income in low-tax foreign affiliates. Finally, the participation exemption reduces the tax cost of repatriating certain foreign earnings, making it easier for firms to reorganize internal multinational structures and redeploy cash within the group in response to the incentives created by the rate cut, FDII, and GILTI. Taken together, these provisions are designed in part to improve the relative competitive position of U.S. firms, but whether they have significant effects on real exports and imports is not obvious *ex ante*.⁴

Because the tax reform represents a common national shock to U.S.-incorporated firms, we employ a quasi-difference-in-differences design that uses predetermined cross-sectional variation in firms' exposure to the reform's international provisions. We measure exposure based on the firms' pre-reform engagement in outbound profit-shifting. Firms engaging in outbound income-shifting before the reform should have been more exposed to the international provisions because the reform changed the tax incentives for locating income and organizing cross-border transactions through the lower corporate tax rate, FDII, and GILTI. Accordingly, our design isolates variation in import and export activity associated with differential exposure to these provisions. These firms were also more likely to have the global subsidiary networks and tax-planning structures needed to adjust cross-border transactions after the reform. In contrast, firms with little pre-reform outbound profit shifting were less directly affected by the international provisions and likely faced higher fixed costs of changing their trade patterns. This design helps mitigate concerns related to omitted variables and

⁴ Firms may also respond by changing intangible asset ownership, internal licensing or transfer-pricing arrangements, or the legal roles of foreign subsidiaries. These responses may occur alongside changes in physical goods trade but could attenuate the observed trade response.

endogeneity and is widely used in tax policy research involving common policy shocks (Albuquerque, Bennett, Lisowsky, and Wang 2025; Amberger and Robinson 2023; Dharmapala, Foley, and Forbes 2011; Dyreng, Gaertner, Hoopes, and Vernon 2023).

We follow prior research to identify firms that shift income out of the United States to foreign jurisdictions during the pre-reform period (Collins, Kemsley, and Lang 1998; Klassen and Laplante 2012; De Simone, Klassen, and Seidman 2022). We classify firms as outbound income shifters if they engage in outbound income shifting during any year from 2014 to 2016. These firms constitute the high-exposure (i.e., treatment) group, while firms with low pre-reform outbound-income-shifting activity serve as the low-exposure (i.e., control) group. To improve comparability, we require treatment and control firms to operate within the same Fama-French 49 industry and to have similar levels of foreign operations prior to the reform. For each firm, we use shipping container data to track its exports to each jurisdiction (i.e., export-relationship) and imports from each jurisdiction (i.e., import-relationship) from three years before to three years after the TCJA enactment. Our final sample includes 10,754 export-relationship-year observations and 45,564 import-relationship-year observations.

Our main analysis examines the effect of the TCJA on firm-level exports and imports separately. We find that firms with high exposure to the law's international provisions increase physical goods exports by about 20.0% and imports by about 13.4% relative to low-exposure firms.⁵ Year-by-year analyses reveal no differential pre-trends, supporting the parallel trends assumption. The increases in exports and imports by U.S. firms occur only in the post-reform period and persist over time, with stronger effects observed in 2019 and 2020, roughly two years after enactment. Throughout the main analysis, we exclude trade with China to mitigate

⁵ Export expansion may require imported intermediate inputs, and multinational firms may reorganize sourcing or internal shipments within global supply chains in response to the reform (Amiti and Konings 2007; Goldberg, Khandelwal, Pavcnik, and Topalova 2010; Kasahara and Rodrigue 2008).

potential confounding effects from the 2018 U.S.-China trade war and associated tariffs.⁶

Robustness tests confirm that our findings are similar under alternative income-shifting definitions, Poisson regressions, alternative trade measures, including weight and dollar value, and an expanded sample that includes trade with China.

We next conduct cross-sectional tests to assess whether the observed trade responses are consistent with tax-based incentives. More profitable firms should have stronger incentives to respond because the value of tax savings is larger when more income is at stake. Firms with greater intangible intensity should also respond more because the TCJA's international provisions, especially FDII and GILTI, were particularly relevant for firms with mobile and intangible assets. We also expect stronger responses when non-tax adjustment costs, such as shipping costs, are lower. Consistent with these predictions, the trade effects are stronger among firms with higher pre-tax profitability, greater R&D intensity, and lower shipping costs. These findings support a tax-based interpretation of the main results.

After documenting that high-exposure firms increased both exports and imports, we next examine the mechanisms underlying these trade responses. We conduct three sets of tests. First, we test whether higher imports reflect substitution away from domestic production. Using domestic facility emissions and production-related job postings, we find no evidence that high-exposure firms reduce U.S. production activity after the reform. Second, we examine whether firms reorganize transactions within multinational networks. We find no decline in vertical integration, but we find an increase in intersegment sales, especially across geographic segments, suggesting greater internal reallocation of transactions across multinational operations. Third, we test whether firms shift toward higher-value or more intangible-intensive

⁶ Therefore, the observed increases in the treated firms' exports and imports are unlikely to be explained by tariff effects, as tariffs typically exert downward pressure on trade volumes. For example, the existing literature that examines the impact of the 2018 U.S. tariffs during Donald Trump's first presidency finds that import and retaliatory tariffs led to significant declines in both exports and imports (Fajgelbaum, Goldberg, Kennedy, and Khandelwal 2020; Handley, Kamal, and Monarch 2024).

products by examining export and import unit prices. We find no systematic changes in unit prices, providing little evidence of product upgrading. Overall, these results suggest that the trade responses do not appear to be accompanied by offshoring of domestic production or systematic changes in traded-product prices. Our findings are consistent with U.S. firms expanding or reorganizing existing cross-border transactions within multinational networks.⁷

Our study makes several contributions to the literature. First, our study contributes to research on the real effects of taxation. Prior studies primarily examine how corporate income tax affects real responses such as investment and employment (e.g., Lester 2019; Fox, Jacob, Wilde, and Wilson 2022; Chen, De Simone, Hanlon, and Lester 2023; Amberger, Gallemore, and Wilde 2026). In contrast, little is known about whether taxes affect firms' physical trade in goods (Dyreng and Maydew 2018; Jacob 2022; Hanlon 2025; Lester and Olbert 2025). This question is difficult to study using conventional accounting data because such measures often combine price and quantity effects and can therefore blur the distinction between nominal and real responses (Glaeser and Lang 2024). This identification problem applies equally to trade outcomes that are measured in financial terms (e.g., sales), which may also reflect transfer pricing decisions and the contractual allocation of transactions across entities rather than the underlying movement of goods.⁸ As Desai and Hines (2001) and others note, firm-level reported foreign sales can be shifted across affiliates through invoicing or licensing arrangements even when physical trade is unchanged.⁹ By using shipment-level quantity data,

⁷ For example, a U.S. parent could export more to a foreign affiliate and use it more intensively as a regional distribution hub, while importing additional complementary inputs. This type of reorganization would increase export and import volumes and geographic intersegment sales through transaction routing, without necessarily changing export or import prices.

⁸ Extant research examines how cross-jurisdictional differences in statutory tax rates affect multinationals' income-shifting behavior (Klassen, Lang, and Wolfson 1993; Hines and Rice 1994; Collins et al. 1998; Mills and Newberry 2004; Klassen and Laplante 2012; Dyreng and Markle 2016; De Simone 2016). For a comprehensive review, see De Simone and Klassen (2026).

⁹ For example, before the TCJA, a U.S. firm selling into Europe might route sales through an affiliate in a low-tax jurisdiction: the affiliate would purchase products at a low transfer price from the United States (or another production location) and then resell them to final customers at a higher price, recording most of the profit abroad. After the TCJA, the firm could have the U.S. parent invoice customers directly and record the associated profit, while keeping the production location and shipping patterns unchanged. In firm-level accounting data, this

we observe firms’ actual cross-border movement of goods more directly than studies based on reported sales or accounting outcomes. To our knowledge, we are the first to examine how this major U.S. tax reform affected firms’ physical trade flows.

Second, our paper advances our understanding of how taxes shape global supply chains. Prior research shows that multinationals respond to cross-country tax differences by shifting income across jurisdictions (De Simone, Mills, and Stomberg 2019; De Vito, Hillmann, Jacob, and Vossebürger 2025; De Simone and Klassen 2026; De Simone and Olbert 2026). We extend this literature by showing that changes in the taxation of multinational income affect not only where income is reported, but also how multinational firms organize cross-border transactions. Our findings indicate that high-exposure firms increase intersegment sales across geographic segments, while domestic production activity and vertical integration remain stable. These results suggest that tax policy can reshape global value chains through internal network reorganization, without necessarily changing where firms produce or from whom they source inputs. Our study, therefore, contributes to the literature on how income tax incentives affect supply-chain configurations (Dyreng, Lindsey, Markle, and Shackelford 2015; Cen, Maydew, Zhang, and Zuo 2017; Chow et al. 2026).¹⁰ We extend this literature by showing that a major tax reform affects firms’ physical cross-border trade flows and internal multinational transactions, highlighting a channel through which tax policy can influence how supply chains operate (Elliott, Golub, and Leduc 2022; Grossman, Helpman, and Lhuillier 2023).

Finally, we contribute to research on the relationship between tax policy and international trade. Much of the existing literature on taxes and trade examines border or

restructuring would appear as an increase in U.S. “export sales” even though real cross-border shipments are unchanged.

¹⁰ Chow et al. (2026) suggest that some *intrafirm* tangible goods imports are “tax driven,” serving as a tool to shift income to lower-tax jurisdictions. Our paper instead examines the broader effect of U.S. tax reform on international trade. Our mechanism does not hinge on tax planning and is not restricted to *intrafirm* trade; we study trade between both related (*intrafirm*) and unrelated parties (*interfirm* trades). As discussed in Section 5.5, our results are similar after controlling for the potential effect of TIEA adoption on trade in a robustness test.

transaction-based taxes, such as tariffs and value-added taxes, which affect trade flows more directly through import or export prices at the border (e.g., Feldstein and Krugman 1990; Slemrod and Velayudhan 2022; Costinot and Werning 2025).¹¹ Much less is known about whether corporate income tax policy affects firms' real trade behavior. This distinction is important because corporate income taxes do not tax trade flows directly. They affect exports and imports through more indirect channels, including the after-tax returns to locating income, production activities, and intangible assets across jurisdictions. One exception is Desai and Hines (2001), which describes the history of U.S. income tax provisions that operated as export subsidies from the 1930s through the 1990s. We extend this literature by examining whether a major modern tax reform, the TCJA, affected firms' physical goods exports and imports. Understanding whether corporate income tax policy affects trade flows is important because, given the scale of U.S. trade, even modest tax-induced changes in firms' cross-border activity could have meaningful aggregate effects. This question is especially salient in the United States, where trade imbalances remain a persistent policy concern.¹²

2. Related Research and Hypothesis Development

2.1 Tax Policy, Real Corporate Responses, and the TCJA

Researchers are increasingly interested in how taxes affect firms' real decisions (Dyreng and Maydew 2018; Jacob 2022; Lester and Olbert 2025). Recent studies investigate whether tax incentives and tax policy changes influence investment, employment, offshoring,

¹¹ Existing research also examines how tariffs influence trade price manipulation. For example, Fisman and Wei (2004) suggest that tariffs and value-added tax on imports incentivize firms to underreport import values so they can evade taxes, which results in a positive correlation between taxes and the discrepancy between reported exports and imports. Javorcik and Narciso (2008) find that tariff evasion primarily involves misrepresenting import prices rather than underreporting quantities or misclassifying products. Beverelli and Ticku (2022) find that trade facilitation measures implemented through WTO Trade Facilitation Agreements significantly reduce the misreporting of trade prices.

¹² For 2025, the trade deficit for goods and services was \$901.5 billion (U.S. Bureau of Economic Analysis 2026).

and the location of economic activity.¹³ For example, Williams (2018) shows that multinational tax incentives affect offshored U.S. jobs, while Lester (2019) finds that domestic production incentives increase investment for some firms but do not necessarily increase domestic employment. Other studies show that innovation box regimes affect capital investment and workforce composition (Chen et al. 2023), that uncertainty about future tax enforcement reduces investment and input purchases (Fox et al. 2022), and that private country-by-country reporting changes multinational firms' investment responses (De Simone and Olbert 2022). Recent work further shows that firms' real responses to tax policy depend on expectations about future tax policy (Gallemore, Hollander, Jacob, and Zheng 2025), tax system complexity (Amberger et al. 2026), and the design of destination-based tax systems (De Simone and Olbert 2026). Collectively, this literature provides evidence that taxes can affect real corporate behavior, but it focuses primarily on investment, employment, location, and organizational decisions rather than on firms' physical cross-border movement of goods.

The TCJA provides a particularly important setting for examining whether tax policy affects physical trade flows because the reform substantially changed the taxation of U.S. multinational income. Since the TCJA's enactment, researchers have examined its effects on firms' tax burdens, avoidance, investments, and financial policies. Much of this literature focuses on how the reform changed corporate tax liabilities and real investment behavior, while leaving its impact on firms' trade flows largely unexplored.

Studies of tax burdens using public-firm data and corporate tax returns (Dobridge, Kennedy, Landefeld, and Mortenson 2023; Dyreng et al. 2023) document that the tax reform lowered effective tax rates on domestic income for both domestic and multinational firms. For multinationals, these tax savings mainly stemmed from reductions in taxes on their domestic

¹³ The effects of taxes on corporate investment location have been studied in a variety of empirical settings (e.g., Hines and Rice 1994; Cummins, Hassett and Hubbard 1996; Chow, Huang, Klassen, and Ng 2022).

earnings rather than foreign earnings; even among multinationals likely subject to anti-abuse provisions, the federal tax burden on foreign income did not change significantly.

Other work examines changes in tax avoidance behavior. García-Bernardo, Janský, and Zucman (2023) show that U.S. firms report modestly less profit abroad, largely due to intellectual property repatriations, while haven profits remain relatively unchanged. Krull and Wu (2023) find that income shifting declines among firms with pre-reform tax rates between 21% and 35%, but not for firms outside that range. Firms also responded to the Base Erosion and Anti-Abuse Tax (BEAT) by reclassifying related-party payments as cost of goods sold (Kelley, Lewellen, Lynch, and Samuel 2024).

Evidence on real investment responses is mixed. Some studies indicate that foreign expansion slowed: foreign acquisitions declined, especially for low-tax targets, consistent with GILTI reducing the value of excess foreign returns (Amberger and Robinson 2023). U.S. firms cut investment in foreign subsidiaries (Samuel 2024) and used repatriated cash primarily for stock buybacks rather than capital investment, M&A, R&D, or employment (Albertus, Glover, and Levine 2025; Albuquerque et al. 2025).¹⁴ In contrast, Chodorow-Reich, Smith, Zidar, and Zwick (2024) find that the tax reform stimulated both domestic and foreign investments. Huang, Osswald, and Wilson (2023) find that FDII encourages U.S. R&D-intensive multinationals to expand their human capital both at home and abroad, suggesting that the provision affected firms' operating structures rather than only the location of reported income.¹⁵

To our knowledge, whether the U.S. tax reform affected U.S. firms' cross-border trade is unknown. We fill this gap by providing the first empirical examination of how the reform's

¹⁴ Carrizosa, Gaertner, and Lynch (2023) document that firms reduced long-term domestic debt following the TCJA's interest-deduction limits, including among firms not immediately constrained, consistent with precautionary deleveraging.

¹⁵ Existing research on Opportunity Zones (OZ) shows mixed real-effects evidence. Arefeva, Davis, Ghent, and Park (2024) find sustained employment growth within OZs, primarily benefiting non-residents, while Corinth and Feldman (2023) conclude the policy did not materially stimulate investment or residents' well-being.

changes to the taxation of multinational income affected U.S. firms' cross-border trade in physical goods.

2.2 Hypothesis Development

The TCJA tax reform introduces several major changes to the U.S. international tax regime that could affect how multinational firms organize cross-border activity and physical goods trade flows. In this setting, the most relevant provisions are those that increase the relative after-tax return to serving foreign markets from the United States rather than through low-tax foreign affiliates. In particular, the reform reduced the tax benefit of locating mobile income offshore while increasing the attractiveness of earning foreign-market income from U.S.-based operations.

Four features of the reform are central to the incentives for U.S. firms' cross-border physical trade. First, the reduction in the U.S. federal corporate tax rate from 35% to 21% narrowed the tax wedge between domestic and foreign income, thereby reducing the tax benefit of locating profits outside the United States.

Second, the FDII provision is, for U.S. exporters, the most direct and favorable element of the post-reform regime. It provides a 37.5 percent deduction for “foreign-derived intangible income,” that is, income earned from U.S.-based intangible assets used to serve foreign markets. The FDII operates in some respects like an export-subsidy (Dharmapala 2018; Clausing 2024).¹⁶ It rewards U.S. firms that keep high-value, IP-intensive operations domestically, making the United States a tax-advantaged place to locate the IP, R&D, and marketing functions that support global sales. By increasing the after-tax return to serving foreign demand from the United States, FDII encourages U.S. firms to retain or bring back

¹⁶ On May 6, 2019, the European Commission submitted comments on the proposed FDII, noting that its design appears likely to meet the criteria of a prohibited export subsidy under the World Trade Organization (WTO) rules (Dombrovskis, Katainen, Malmstrom, and Moscovici 2019). The U.S. Congressional Research Service shares this perspective, stating in its 2021 report that the FDII “would probably violate the WTO rules against export subsidies” (Congressional Research Service 2021).

intangible assets and export-supporting functions to the U.S., which could increase physical goods exports (Polantz 2020; Urban-Brookings Tax Policy Center 2024b).

Third, GILTI works in tandem with FDII to make the United States a more attractive base for export operations by penalizing the offshore holding of IP. GILTI is designed to tax “mobile” income, primarily profits from intangible assets, such as patents and trademarks, that are earned by controlled foreign corporations in low-tax jurisdictions. It imposes a U.S. minimum tax on foreign income in excess of a 10 percent return on foreign tangible assets, reducing the tax advantage of reporting income in low-tax jurisdictions and weakening incentives to allocate high-margin activities abroad.

Conceptually, GILTI complements FDII: FDII makes it more attractive to hold intangibles in the United States and use them to serve foreign customers, while GILTI makes it less attractive to keep those assets offshore. In this sense, FDII functions as the “carrot” for locating IP domestically, while GILTI operates as the “stick” discouraging the location of intangibles abroad. Together, the two provisions encourage firms to locate IP and export-related profits in the United States: under FDII, firms face a lower effective tax rate on export-related income derived from U.S. intangibles, whereas under GILTI, intangible-related income earned overseas is subject to additional U.S. tax.¹⁷

Finally, the participation exemption, implemented through a 100 percent deduction for certain foreign-source dividends, moves the U.S. system in a more territorial direction by largely eliminating residual U.S. tax on qualifying foreign dividends. By reducing the tax cost of repatriating foreign earnings, this change allows U.S. multinationals to redeploy foreign profits more freely within the firm, potentially improving internal capital market efficiency (Blouin, Krull, and Robinson 2023) and supporting domestic investment (Hanlon,

¹⁷ The FDII deduction implies a reduced tax rate of 13.125 percent $[(1-37.5%)*21\%]$ on qualifying income, while GILTI corresponds to an additional minimum U.S. tax rate of about 10.5–13.125 percent on relevant foreign income, depending on the availability of foreign tax credits.

Lester, and Verdi 2015; Edwards, Kravet, and Wilson 2016). The provision may also facilitate internal structuring, such as unwinding holding-company structures, as firms adjust to the new incentives created by the rate cut, FDII, and GILTI.

Taken together, these provisions increase firms' tax incentives to earn foreign-market income through their U.S.-based entities rather than through low-tax foreign affiliates. This yields the central prediction that firms with greater exposure to the reform's international provisions should expand physical goods exports from the United States.¹⁸

We also examine imports because multinational firms often organize production across global value chains. When firms increase exports, they may also increase imports of intermediate inputs and components used in production or distribution (Kano, Tsang, and Yeung 2020; Antràs and Chor 2022).¹⁹ Thus, studying exports and imports together provides a more complete picture of how the tax reform affected firms' cross-border physical trade. The implications for imports are more nuanced, but they can be integrated with the same underlying mechanism. The literature on global production networks and imported intermediates suggests that exporting and importing are often complementary activities: firms that expand foreign sales commonly increase imports of inputs or reorganize sourcing across borders to support those sales (Amiti and Konings 2007; Goldberg et al. 2010; Kasahara and Rodrigue 2008). Accordingly, if the tax reform increases the attractiveness of serving foreign markets from U.S.-based entities, more exposed firms should also increase imports of intermediates, components, and related-party shipments that complement U.S.-based production and foreign sales. Imports may therefore increase because expanded export

¹⁸ We proxy exposure using the engagement in outbound profit shifting prior to the reform. This measure captures where the reform is most likely to materially change after-tax returns to domestic versus foreign production and to different cross-border trade configurations, and it also indicates that treated firms already operate established international trade and logistics networks, making short-run adjustments in global trade more feasible. Throughout the analysis, we refer to firms with high pre-reform outbound income shifting as having "higher exposure" to the international provisions and firms with low or no shifting as having "lower exposure."

¹⁹ According to S&P Global, 43.8% of U.S. imports carried by liner shipping in 2023, valued at approximately \$490 billion, were intermediate inputs used by American businesses in production processes, including components, supplies, and raw materials (Bingham et al. 2025).

activity requires complementary imported inputs or because firms reorganize intra-firm transactions within multinational production networks. Thus, while exports represent the more direct response to the reform's international tax incentives, imports may also increase as a complementary response.²⁰

These considerations lead to the following hypotheses, stated in alternative form:

H1: *After the enactment of the TCJA, U.S. firms with higher exposure to the tax reform's international provisions will increase physical goods exports relative to firms with lower exposure.*

H2: *After the enactment of the TCJA, U.S. firms with higher exposure to the tax reform's international provisions will increase physical goods imports relative to firms with lower exposure.*

However, it is important to note that the reform's international tax incentives need not translate into changes in cross-border physical goods trade. First, the participation exemption may reduce the need for U.S. firms to alter existing offshore structures and may have made foreign affiliate operations easier to maintain by reducing the tax cost of repatriating foreign earnings. Second, prior evidence suggests that repatriation tax relief can lead firms to return cash to shareholders rather than expand real activity, particularly through share repurchases (Blouin and Krull 2009; Albuquerque et al. 2025). Third, firms may respond partly by changing the ownership of intangible assets, revising transfer pricing arrangements, or modifying the roles of foreign subsidiaries within the multinational group. These adjustments can affect where income is reported without necessarily requiring proportional changes in firms' physical goods export and import flows. Such responses may occur alongside changes in physical trade, but they could attenuate the magnitude of the observed trade response. Thus, whether the reform's

²⁰ Our shipment data do not allow us to distinguish intrafirm from arm's-length trade or intermediates from final goods, so our empirical tests focus on the net effect on firms' aggregate exports and imports.

international tax provisions translated into measurable changes in physical goods trade is ultimately an empirical question.²¹

3. Sample and Research Design

3.1. Sample Construction

We access data pertaining to U.S. companies' exports and imports from the years 2007 through 2020 from the S&P Global Panjiva database. In accordance with Title 19 of the U.S. Code of Federal Regulations (CFR), U.S. companies are required to furnish comprehensive shipment particulars in the cargo declarations they submit to the U.S. Customs and Border Protection (CBP). For each shipment transaction, Panjiva compiles an extensive set of data, including the senders, the consignee, the shipments' origin and destination, the product codes and descriptions of the items shipped, etc.²²

Our analyses focus on international trade between non-U.S. jurisdictions and U.S. corporations. For the export analyses, we refer to a U.S. firm-destination jurisdictions as export trade relationships. For the import analyses, we create a relationship-year panel with each relationship referring to U.S. firm-origin jurisdictions as import trade relationships. We then create an annual export (import) relationship panel that spans from 2007 to 2020. We conduct the analyses at the annual level to mitigate concerns related to the large seasonal fluctuations in international trade. We exclude trade relationships with China in order to avoid the

²¹ FDII's complex rules on eligible income, deemed intangible income, foreign-derived income, and documentation may raise compliance costs, limiting firms' ability to quickly incorporate the deduction into real or organizational decisions. As prior work shows that tax complexity can reduce the take-up and effectiveness of tax incentives (Cui, Hicks, and Xing 2022), FDII's trade effects may be muted, delayed, or concentrated among firms with stronger tax-planning capacity.

²² A potential concern is that firms can redact supplier identities in the Panjiva database to protect commercially sensitive information. Two features of our setting mitigate this concern. First, redaction almost always conceals a supplier's identity rather than its jurisdiction, and because we aggregate shipments to the firm-jurisdiction level, the loss of individual supplier identifiers has limited impact on our measures. Second, redaction is unlikely to introduce systematic bias. Because the U.S. government retains access to the complete, unredacted records, firms have no incentive to alter their redaction decisions in response to tax policies or enforcement, leaving redaction plausibly uncorrelated with our variables of interest.

confounding effect of the trade war that took place in 2018, although our results are not sensitive to this empirical choice. For each export (import) trade relationship, we aggregate the shipment volume in the twenty-foot equivalent unit (TEU) at the annual level, and we define the transaction volume for one year of a relationship as the total volume in TEU of the containers shipped to (from) a jurisdiction from (to) the U.S. firm in a year. Missing values — a relationship-year without trade in a year — are set to zero.

To investigate the impact of the TCJA tax reform, we restrict the sample period to the years 2014 to 2020, which encompasses the three years before and after the reform’s 2017 enactment. This short event window helps to mitigate concerns about confounding events. To analyze the temporal changes in trading, we further require that a trade relationship have at least one trade record within the sample period. To eliminate the impact of order backlogs from the pre-period, we exclude observations from 2017, the TCJA’s enactment year.

3.2. Identification Strategy

The TCJA tax reform constituted a sharp overhaul of U.S. corporate tax provisions, and it is widely regarded as an exogenous policy shock (Amberger and Robinson 2023; Carrizosa et al. 2023; Wagner, Zeckhauser, and Ziegler 2018). Although some degree of reform was anticipated after the 2016 election, the legislative framework was not released until September 2017, and the bill moved rapidly from proposal (November 2, 2017) to enactment (December 22, 2017) and implementation (January 1, 2018), leaving little scope for anticipatory corporate behavior (Carrizosa et al. 2023; Gaertner, Lynch, and Vernon 2020). Crucially for our context, the tax reform did not prompt major international tax policy responses; key U.S. trading partners largely left their regimes unchanged (Amberger and Robinson 2023; Chalk, Keen, and Perry 2018). This absence of contemporaneous foreign reforms helps to isolate the reform’s economic effects and strengthens the credibility of our inferences.

Because the TCJA represents a common national shock to U.S.-incorporated firms, we employ a quasi-difference-in-differences design that exploits predetermined cross-sectional variation in firms' exposure to the reform's international provisions. This design allows us to focus on variation in treatment intensity arising from firms' pre-existing exposure to the reform, helping to mitigate concerns related to omitted variables and endogeneity. Similar approaches are widely used in tax policy research involving common shocks (e.g., Albuquerque et al. 2025; Amberger and Robinson 2023; Dharmapala et al. 2011; Dyreng et al. 2023).²³

We measure exposure using firms' pre-reform engagement in outbound profit-shifting. Firms engaging in outbound income shifting before the reform should have been more exposed to the international provisions because the reform changed the incentives that had previously encouraged U.S. multinationals to locate income in low-tax foreign jurisdictions. The lower corporate tax rate, the FDII deduction, and the GILTI provision reduced the relative tax benefits of outbound income shifting and increased the attractiveness of earning foreign-market income through U.S.-based operations. Accordingly, firms that had taken greater advantage of the pre-reform international tax system were more likely to experience a meaningful change in incentives after the reform. Our empirical approach isolates variation in export and import activity associated with firms' differential exposure to these provisions. At the same time, these firms are likely to have the global subsidiary networks and tax-planning structures needed to adjust cross-border transactions in response to the reform. In contrast, firms with little pre-reform outbound profit shifting were less directly affected by the reform's international provisions.

²³ Amberger and Robinson (2023) and Dharmapala (2018) measure a firm's exposure to the tax reform and the Homeland Investment Act, respectively, by examining firms' repatriation tax costs. Similarly, Dyreng et al. (2023) focus on the level of permanently reinvested earnings prior to the tax reform.

We follow prior studies to identify firms that shift income out of the United States to foreign jurisdictions in the pre-reform period (Collins et al. 1998; Klassen and Laplante 2012; De Simone et al. 2022). The treatment group, or the high-exposure group, consists of firms identified as outbound income shifters in any year during the pre-reform period from 2014 to 2016. Firms not identified as active outbound income shifters during this period are classified as low-exposure firms or the control group.²⁴ To improve comparability between the treatment and control groups, we require firms to operate in the same Fama-French 49 industry and to have similar levels of foreign operation activities prior to the reform, measured by quartiles of the ratio of foreign income to total assets. After requiring the availability of the control variables used in the main analyses, our final trade sample consists of 10,754 export-relationship-year and 45,564 import-relationship-year observations. Table 1, Panels A and B summarize the sample selection procedures for the export and import samples, respectively. Panel C reports the distribution of the sample firms across Fama-French 49 industries, showing that our sample encompasses a wide range of industries.

3.3. Research Design

To test the effects of the tax reform on international trade, we estimate Equation (1) separately for the export and import relationship samples.

$$Trade_{i,j,t} = \beta_1 High\ Exposure_i \times Post_t + \beta_2 X_{i,j,t-1} + \tau_t + v_{i,j} + e_{i,j,t}, \quad (1)$$

where i , j , and t denote the firm, the jurisdiction, and the year. $Trade_{i,j,t}$ is either *Export Volume* or *Import Volume*. *Export Volume* $_{i,j,t}$ is defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) that firm i exports to jurisdiction j in year t . *Import*

²⁴ Dyreng et al. (2023) classify firms with higher levels of permanently reinvested earnings in 2016 as those facing anti-abuse provisions in the TCJA. They find that multinationals see no change in the federal tax burden on foreign income, even if the firms are subject to anti-abuse provisions. In contrast, firms not targeted by anti-abuse provisions experience decreases in the federal tax burden on their foreign income. Their results alleviate concerns that our observed findings are driven by active outbound income-shifting firms (our high-exposure group) experiencing significant changes in foreign tax burdens.

$Volume_{i,j,t}$ is defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) that firm i imports from jurisdiction j in year t . $High\ Exposure_i$ is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. $Post_t$ is a binary variable that equals one for the years after the reform enactment (i.e., 2018 to 2020). τ_t is the year fixed effects, which enable us to account for macro factors, such as the pandemic years, which may influence international trade. $v_{i,j}$ is the relationship (i.e., firm-jurisdiction) fixed effects, and they allow us to control for time-invariant relationship characteristics (e.g., the distance to the exporting or importing jurisdiction).

In keeping with prior studies (Dyreng et al. 2015; Bennedsen and Zeume 2018; De Simone et al. 2019; Chow et al. 2026), we control for a set of firm-level characteristics. We include the pre-tax returns on assets (ROA_EBITDA), the natural logarithm of the total assets ($Size$), the market-to-book ratio (MTB), the growth of sales ($Sales\ Growth$), and the leverage ratio ($Leverage$) to account for firm performance. Next, we include the following ratios: inventory to total assets ($Inventory$), fixed to total assets (PPE), capital to total assets ($CAPX$), intangible to total assets ($Intang$), and R&D expenditure to total assets (RD), which serve to measure the importance of operating, tangible, and intangible investments. Furthermore, we include the ratio of pre-tax foreign income ($Foreign\ Income$) to control for the importance of foreign sales, and the annual change in the net operating loss carryforward ($Change\ NOL$) and an indicator for a positive net operating loss carryforward (NOL) to control for the tax loss carryforward. Finally, we cluster standard errors by firm-jurisdiction to account for the correlation of trade with a trade relationship.

3.4. Descriptive Statistics

Table 2 reports the summary statistics for the key variables for the export (Panel A) and import samples (Panel B). The average *Export Volume* (*Import Volume*) is 1.185 (0.853), suggesting that the average firm exports 1.482 (= $\sinh(1.185)$) containers (TEU) to a given

foreign jurisdiction, and it imports 0.960 ($=\sinh(0.853)$) containers (TEU) from a given foreign jurisdiction in a year. The sample means appear small because many firm-jurisdiction-years do not have import or export activities.

In the export sample, the average firm has a pre-tax ROA of 15.2%, assets of 8,569 million, a market-to-book ratio of 3.069, and a leverage ratio of 30.7%. On average, inventory, PP&E, and intangible assets account for 13.9%, 26.6%, and 24.0% of total assets, respectively, and the average firm spends 4.4% and 2.1% on capital and R&D expenditure, respectively, relative to total assets. In the import sample, the average firm has a pre-tax ROA of 14.3%, assets of 5,224 million, a market-to-book ratio of 3.868, and a leverage ratio of 24.3%. On average, inventory, PP&E, and intangible assets account for 15.0%, 23.8%, and 24.4% of the total assets, respectively. The average firm respectively spends 4.3% and 2.7% on capital and R&D expenditure relative to its total assets. These statistics suggest that exporters tend to be larger and more mature than importers. They are also largely consistent with prior studies (e.g., Klassen and Laplante 2012; Dyreng et al. 2015; De Simone et al. 2019).

4. Empirical Results

4.1. Main Results

Table 3 reports the estimates from Equation (1), which tests our hypotheses. Panel A reports the results for export volume. In column (1), we estimate Equation (1) without the firm-level controls. Column (2) reports the results after the inclusion of a host of firm characteristics. We find that the coefficients on *High Exposure* $\times$ *Post* are positive and significant in both specifications (0.258; p -value < 0.01 in column (1); 0.200; p -value < 0.01 in column (2)). The latter estimate suggests that the post-reform exports of high-exposure firms increase by about 20.0%.²⁵

²⁵ Following Bellemare and Wichman (2020), coefficients can be interpreted as approximate percentage changes (semi-elasticities) when the dependent variable is sufficiently large.

Panel B examines the impact on the import volume by estimating Equation (1) using *Import Volume* as the dependent variable. The coefficient on *High Exposure* $\times$ *Post* is positive and significant in both specifications. In terms of economic magnitudes, the estimate in Column (2) suggests that following the tax reform, the imports of high-exposure firms increase by 13.4%, which is consistent with our expectation.

Overall, these results support H1 and H2: relative to low-exposure firms, firms that are more exposed to the international provisions increase both their physical goods exports and imports after the tax reform, with a stronger response for exports.

The key identification assumption that underlies the quasi-difference-in-differences design is that, absent the tax reform, the high-exposure firms' trade trends would be the same as those for low-exposure firms. To validate this parallel trend assumption, we replace *High Exposure* $\times$ *Post* with interaction terms between *High Exposure* and the year indicators from the three years before to the three years after the reform enactment. Figure 1 plots the coefficients on the interaction terms and their associated 95% confidence intervals. We omit year $t-1$, which serves as the benchmark period. Panels A and B show estimates for exports and imports, respectively. In both panels, the pre-reform coefficients are statistically indistinguishable from zero, which lends support to the parallel trend assumption. In contrast, the post-reform coefficients are significantly positive, indicating that the tax reform has a persistent positive impact on both exports and imports of physical goods by U.S. firms.

In sum, the results in Table 3 and Figure 1 document that, after the tax reform, firms with greater exposure to the international provisions expand their international trade, increasing both physical goods exports and imports, with exports rising more than imports.

4.2. Cross-sectional Tests

To provide further evidence on the tax-based mechanisms, we examine heterogeneity in the trade response across firms. These tests help assess whether the observed increases in

physical goods exports and imports are consistent with tax incentives, rather than reflecting contemporaneous non-tax shocks. We focus on firm characteristics that, ex ante, affect either the expected benefits of responding to the reform or the non-tax costs of adjusting trade. In particular, we expect stronger trade responses from firms that are (i) more profitable, (ii) more intangible-intensive, and (iii) subjective to lower non-tax adjustment costs, such as shipping costs. We therefore conduct three cross-sectional tests along these dimensions.

4.2.1 Pre-tax Profitability

We first examine heterogeneity in pre-tax profitability. The reform's statutory rate cut and FDII reduce the tax burden on qualifying income earned by U.S. corporations. Because the dollar value of a lower tax rate increases with the size of the income base, the expected benefits of the reform should be greater for more profitable firms. Accordingly, high-exposure firms with higher pre-tax profitability should have stronger incentives to reorganize trade activity in response to the reform, whereas firms with lower profitability may find the expected tax benefits too small to justify costly real adjustments.

To test this prediction, we create a binary variable, *Low Profitability*, to indicate firms with a pre-tax return on assets ratio in the bottom quartile of the sample distribution. We then re-estimate Equation (1) by interacting *Low Profitability* with *High Exposure* $\times$ *Post*. We report the results in Table 4. Column (1) details the results of the export effects. We find that *High Exposure* $\times$ *Post* is significantly positively related to *Export Volume*, suggesting that high-exposure firms with greater pre-tax income increase their exports to derive tax benefits. In contrast, *High Exposure* $\times$ *Post* $\times$ *Low Profitability* is significantly negative, suggesting that the effect of the tax reform is muted for low-profitability firms. Column (2) shows an analogous pattern for imports: *High Exposure* $\times$ *Post* is significantly positive, while *High Exposure* $\times$ *Post* $\times$ *Low Profitability* is significantly negative.

Together, these results indicate that the observed increase in cross-border physical goods trade is concentrated among firms with high pre-tax profitability. This outcome is consistent with our expectations and lends further support to the interpretation that our main results are tax-driven.²⁶

4.2.2 R&D Intensity

We next examine heterogeneity in firms' investment in intangible assets. The FDII provision gives firms a direct incentive to expand their foreign-derived income associated with domestic intangible capital. GILTI, in turn, reduces the tax advantage of earning high-return income through controlled foreign corporations in low-tax jurisdictions. This change in incentive is likely to be more relevant for firms with greater investments in R&D, as such firms are more likely to hold valuable intellectual property, incorporate that intellectual property into goods sold abroad, and earn high-margin income from foreign markets. Accordingly, intangible-intensive firms may have greater scope to respond to the reform by expanding foreign sales supported by U.S.-based activities.

If the reform's intangible-related international tax incentives are an important channel behind our main results, we would expect firms with greater R&D intensity to exhibit larger post-reform changes in their trade volume. To test this prediction, we partition firms based on their R&D intensity, defined as the R&D expense scaled by lagged total assets. We define *Low R&D* as a dummy variable indicating that *R&D Intensity* falls in the bottom quartile of the sample distribution. We then augment Equation (1) with the *Low R&D* indicator and its interactions with *High Exposure* $\times$ *Post*. Table 5 presents the results, with columns (1) and (2) reporting the findings for *Export Volume* and *Import Volume*, respectively.

²⁶ We recognize that the pre-tax profitability test may also capture differences in firms' ability to finance costly organizational changes. In this alternative, non-tax interpretation, firms with low pre-tax profits may be less able to restructure their existing IP arrangements, limiting their capacity to adjust exports in response to the tax reform.

We find that in both columns, *High Exposure* $\times$ *Post* is significantly positive, while *High Exposure* $\times$ *Post* $\times$ *Low R&D* is negative and significant at the 5% level. These findings indicate that the post-reform increase in trade volume is concentrated among firms with higher R&D intensity. Overall, these findings suggest that firms with greater R&D intensity were better positioned to benefit from the TCJA's international tax incentives and, as a result, exhibited stronger adjustments in physical goods trade volumes after the reform.

4.2.3 Transactions Costs

Finally, we examine heterogeneity in the shipping costs for physical goods imports, which represent significant non-tax frictions. Real responses such as trade adjustments can be costly due to additional transaction expenses, including shipping. Rational managers weigh the tax reform's tax benefits against these extra costs and are more likely to adjust trade patterns when incremental shipping costs are relatively low. Accordingly, we posit that high shipping costs may deter adjustments to physical goods trades, reducing firms' responsiveness to the tax reform.

We measure the costs of international trade following the approach of Bernard, Jensen, and Schott (2006) and Barrot, Loualiche, and Sauvagnat (2019). Specifically, we use U.S. import data to construct the industry-year (four-digit SIC) shipping costs as the markup of the Cost-Insurance-Freight (CIF) value over the Free-on-Board (FOB) value. Because shipment cost information is only available for importing activities, this analysis focuses on firms' imports. We then classify firms with industry-year shipping costs below the top quartile of the sample as having low shipping costs, and interact the *Low Import Shipping Cost* indicator with the tax reform exposure treatment variable.

We report the results in Table 6. Note that the sample size is reduced because shipment cost data are only available for manufacturing firms. The coefficient on the interaction term is significantly positive (0.206, $p < 0.05$), indicating that, in the wake of the reform, firms with

lower shipping costs are significantly more likely to increase their physical goods imports. These findings are consistent with our expectation that higher shipping costs diminish firms' trade response to the reform's international provisions.

In summary, we find that the main effects of the tax reform on physical trade volume vary systematically with firms' expected tax benefits and non-tax adjustment costs (i.e., their pre-tax profitability, intangible investment intensity, and shipping costs) in the anticipated directions. These patterns further support the interpretation that the reform's international tax incentives are the primary driver of the observed trade responses.

4.3. Changes in Real Production and Multinational Network Reorganization

Our main results show that, after the reform, firms more exposed to the TCJA's international provisions increase both physical goods exports and imports relative to less-exposed firms. However, the observed higher imports may complement U.S.-based export activity, for example by providing intermediate inputs for domestic production. Alternatively, higher imports may reflect substitution away from U.S. production toward foreign production or sourcing. The increase in shipment volumes may also reflect a reorganization of transactions within multinational networks. In addition, the reform may affect the composition or unit value of traded goods if firms shift toward higher-value or more intangible-intensive products. We therefore conduct three sets of mechanism tests that distinguish among these potential adjustments.

4.3.1 Domestic Production Substitution

We first test whether the import response reflects substitution away from U.S. production. Although our main results show a significant increase in exports, firms may offshore particular stages of production while continuing to ship goods from the United States. FDII provides a specific reason this channel may matter, because the deduction applies to foreign-derived

income in excess of a deemed normal return on domestic tangible assets, creating an incentive to reduce the U.S. tangible asset base (Dharmapala 2018; Clausing 2024).²⁷ If exposed firms are replacing domestic production with foreign production, we should observe weaker indicators of U.S. production activity.

As comprehensive firm-level measures of aggregate domestic production are unavailable, we employ two complementary proxies that capture different aspects of production activity. The first proxy is the total toxic emissions reported to the EPA’s TRI database, which provides a facility-level proxy for domestic production intensity. The second proxy is the total number of production-related job postings from the Lightcast database, which captures firms’ demand for domestic production labor. We conduct the analysis using a firm-year panel covering all firms included in either the import or export sample. Specifically, we estimate the following difference-in-differences regression at the firm-year level:

$$Y_{i,t} = \beta_1 High\ Exposure_i \times Post_t + \beta_2 X_{i,t-1} + \tau_t + v_i + e_{i,t}, \quad (2)$$

where i and t denote the firm and year, respectively. *High Exposure* is an indicator for firms classified as pre-reform outbound income shifters, and *Post* equals one for post-2017 years. All regressions include firm and year fixed effects.

Panel A of Table 7 reports the results. In column (1), the dependent variable is the natural logarithm of total emissions from all facilities in the U.S. *High Exposure* $\times$ *Post* is statistically insignificant. Column (2) uses the inverse hyperbolic sine of the total number of online job postings that the firm issues during the year as the dependent variable, and *High Exposure* $\times$ *Post* continues to be statistically indistinguishable from zero. These results suggest that high-exposure firms do not scale down their domestic production following the tax reform.

²⁷ This incentive is potentially economically meaningful given the large scale of FDII claims and associated revenue losses. U.S. Joint Committee on Taxation (2017) projected a \$64 billion loss in tax revenue over the 10-year period from 2018 to 2027. In comparison, data from the Penn Wharton Budget Model (2023), as discussed by Clausing (2024), indicates that the foregone federal corporate tax revenue is roughly \$50 billion over the period 2018–2020. For context, U.S. corporations claimed \$53 billion in FDII deductions in 2018, \$65 billion in 2019, and \$71 billion in 2020 (Penn Wharton Budget Model 2023).

To further assess whether firms selectively outsource their lower-value-added activities, we decompose job postings into high- and low-skill categories. If firms primarily offshore routine or low-value tasks, we would expect a decline in low-skill postings. However, Columns (3) and (4) show that job demand for both high- and low-skill workers remains unchanged. Overall, these results suggest that domestic production activity remains stable following the reform, which weighs against an interpretation in which the increase in imports reflects large-scale offshoring of tangible production.

4.3.2 Multinational Network Reorganization

Second, we examine whether firms alter the boundaries and internal organization of production within multinational networks. If firms replace internal production with arm's-length offshore suppliers, we should observe declines in vertical integration or intra-firm production linkages. By contrast, if the reform leads firms to reorganize transactions within multinational networks, trade may increase without a decrease in domestic production, and intersegment sales may increase across business or geographic segments.

Following Bourveau, Kepler, She, and Wang (2024), we employ two measures of vertical integration. The first is the vertical relatedness of firms' products disclosed in annual reports, as developed by Frésard, Hoberg, and Phillips (2020). The second is the volume of intersegment sales reported on Compustat segment data, which captures the revenues generated from sales to other business or geographic segments within a firm. We estimate the difference-in-differences regression at the firm-year level using Equation (2).

As reported in Table 7, Panel B, we find no evidence that vertical integration declines for high-exposure firms following the reform. However, the coefficient on *High Exposure* $\times$ *Post* is significantly positive for total intersegment sales as shown in column (2), suggesting an increase in intra-firm transactions among high-exposure firms. In columns (3) and (4), we decompose intersegment sales into sales between business segments (i.e., segments with

different production foci) and sales between geographic segments (i.e., segments located in different geographic markets). We find that the increase is mainly driven by sales between geographic segments.

This result provides evidence of expanded intra-firm transactions across geographic segments and weighs against the notion that exposed firms dismantle internal production chains and replace them with arm's-length offshore sourcing. Combined with the absence of a decline in domestic production proxies in Panel A, this evidence suggests that the increase in shipment volumes occurs without a corresponding contraction in domestic real activity. The findings indicate that reorganization within multinational firm networks is an important margin through which high-exposure firms adjust following the reform.²⁸

4.3.3 Product Upgrading

Third, we test whether firms shift toward higher-value, more intangible-intensive products. If exposed firms retain domestic production but upgrade their product mix, they may import relatively lower-value tangible inputs and export relatively higher-value outputs. This channel predicts changes in import and export unit prices. We therefore examine whether the reform is associated with changes in export and import unit prices.

To test this product-upgrading channel, we re-estimate Equation (1) using the unit price of exports or imports as the dependent variable. We measure unit value either as the price per kilogram or price per unit volume. As Table 8 shows, the interaction term *High Exposure* $\times$ *Post* is statistically insignificant across all four specifications. This result suggests that, relative

²⁸ Firms with existing global networks could respond by exporting more from the United States, importing more complementary inputs, and routing more transactions through foreign distribution or procurement affiliates. For example, a U.S. parent may export more goods to an existing or newly established European affiliate and use that affiliate more intensively as an internal distribution hub for Germany, France, and Spain. Because this response operates through transaction volume and geographic routing, it can generate higher export volumes, modestly higher import volumes, and higher geographic intersegment sales without changing export or import prices.

to low-exposure firms, high-exposure firms do not exhibit systematic changes in either the export or import unit prices following the reform.

These findings provide little support for the product-upgrading explanation. The absence of systematic changes in unit prices is consistent with firms increasing shipment volumes within existing product lines, without substantial changes in product mix or unit prices. It is important to note, however, that unit-price analyses are based on a substantially smaller sample because many shipments do not report the transaction values. As such, these tests should be interpreted with caution.

5. Robustness Tests

5.1. Alternative Identification Strategy

In our primary analysis, we identify firms as income shifters if they shifted their income during any pre-period year (2014–2016); these firms constitute the treatment group. To assess the sensitivity of our results to this approach, we conduct a robustness check where the treatment group includes firms that consistently shifted income across all the pre-period years. We then re-construct the regression sample following the procedures in Section 3 and re-estimate Equation (1) by replacing the indicator variable *High Exposure* with *High Exposure_alt*, which corresponds to this new treatment group. Panel A of Table 9 shows that *High Exposure_alt* $\times$ *Post* is positively associated with both *Export Volume* (0.405, $p < 0.01$) and *Import Volume* (0.216, $p < 0.01$). The magnitude of these effects exceeds that reported in Table 3, consistent with our expectation that this alternative approach captures firms with the strongest incentives and capacity for income shifting.

5.2. Poisson Regression

In Table 9, Panel B, we re-estimate our main specifications from Table 3 using Poisson regressions to address concerns that a zero or near-zero value in the trade data might bias our

findings (Cohn, Liu, and Wardlaw 2022). Instead of logged dependent variables, we use the raw, unlogged values of the export and import volumes. Columns (1) and (2) respectively present the results for export and import effects. The coefficients on *High Exposure* $\times$ *Post* are positive and statistically significant (0.349, $p < 0.05$ for exports; 0.275, $p < 0.01$ for imports), and their magnitudes are larger than those in Table 3, reinforcing the robustness of our results.

5.3. Alternative Trade Volume Measures

In the third robustness check, we employ alternative measures of trade volume, namely, the container weight and the dollar value of trade. The results are reported in Panel C of Table 9. Columns (1) and (2) show the container weight results, while columns (3) and (4) report the results using the dollar value. Across all measures, *High Exposure* $\times$ *Post* remains significantly positive, indicating that following the tax reform, high-exposure firms experienced increases in both the weight and dollar value of their exports and imports.

5.4. Including Trade with China

Throughout the main analysis, we exclude trade with China to mitigate potential confounding effects from the 2018 U.S.-China trade war and tariffs. To test the robustness of our findings, we include trade relationships with China. As shown in Table 9, Panel D, the coefficients on *High Exposure* $\times$ *Post* remain both economically and statistically similar, suggesting that our results are not materially driven by the trade war dynamics with China.

5.5. Controlling for Bilateral Tax Information Exchange with the United States

Using the signing of Tax Information Exchange Agreements with the United States as a proxy for increased tax enforcement, Chow et al. (2026) find that, after the United States enters into a TIEA with a given jurisdiction, U.S. firms significantly reduce their imports from that jurisdiction. To ensure that our estimates are not confounded by TIEA adoption during our sample period, we augment Equation (1) with the interaction term *TIEA* $\times$ *Post_TIEA*, where

TIEA is an indicator for jurisdictions that enact a *TIEA* with the United States during the sample period, and *Post_TIEA* is an indicator for the post-TIEA period. We include the interaction $TIEA \times Post_TIEA$ to capture the differential change in trade with TIEA jurisdictions after adoption. Panel E of Table 9 shows that *High Exposure* $\times$ *Post* remains positively associated with both *Export Volume* (0.200, $p < 0.01$) and *Import Volume* (0.134, $p < 0.01$). The magnitudes of these effects are comparable to those reported in Table 3, suggesting that our results are not driven by jurisdictions that adopted a bilateral TIEA with the United States during our sample period.

6. Conclusion

In this study, we examine the trade effects of the 2017 U.S. tax reform using detailed firm-jurisdiction-level shipping data on physical containerized trade flows from 2014 to 2020. To the best of our knowledge, no prior study directly examines firm-level trade responses to this tax reform. By using shipment-based measures of quantities and routes, we observe changes in firms' physical goods exports and imports, largely insulated from contractual reallocations of sales or profits across affiliates and from financial statement measures that combine both price and quantity. Our study thus fills an important gap in understanding how the tax reform affected U.S. firms' real international trade activity.

Using a quasi-difference-in-differences design, we find that, after the reform, firms that are more exposed to the international provisions of the tax reform significantly increase both their exports and imports. These findings suggest that, as a comprehensive package, the reform is associated with a significant increase in physical goods trade among U.S. firms more exposed to the international provisions. Cross-sectional analyses further show that these trade responses are stronger where expected tax benefits are larger and adjustment costs are lower, consistent with tax incentives shaping firms' real trade behavior.

We also examine mechanisms underlying these trade responses. The evidence points to expanded transactions within multinational networks, particularly across geographic segments, after the reform. Additional tests show that higher imports are not accompanied by a contraction in U.S. production activity. Export and import unit prices also do not change systematically, providing little evidence of broad product upgrading or changes in traded-product values. Overall, these findings suggest that the reform affected the organization and scale of firms' existing multinational transactions, with little evidence of offshoring domestic production or systematic changes in product prices.

Taken together, our findings suggest that the tax reform is associated with increased physical goods trade volumes among U.S. firms more exposed to the international provisions. These findings are informative for policymakers, as they suggest that the reform may have strengthened U.S. firms' global engagement while preserving domestic production. They also highlight the usefulness of physical shipment data for evaluating tax and trade policies, especially when such policies affect both where firms undertake real activity and where they report income for tax purposes.

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Appendix I Variable Definition

Variable	Definition
<i>Dependent Variables</i>	
<i>Export Volume</i>	The inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped to a given foreign jurisdiction from the firm during the year.
<i>Import Volume</i>	The inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year.
<i>Export Raw</i>	The raw volume of the containers (measured in TEU) shipped to a given foreign jurisdiction from the firm during the year.
<i>Import Raw</i>	The raw volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year.
<i>Export Weight</i>	The inverse hyperbolic sine of the weight of the containers (measured in kilograms) shipped to a given foreign jurisdiction from the firm during the year.
<i>Import Weight</i>	The inverse hyperbolic sine of the weight of the containers (measured in kilograms) shipped from a given foreign jurisdiction to the firm during the year.
<i>Export Value</i>	The inverse hyperbolic sine of the value of the containers (measured in USD) shipped to a given foreign jurisdiction from the firm during the year.
<i>Import Value</i>	The inverse hyperbolic sine of the value of the containers (measured in USD) shipped from a given foreign jurisdiction to the firm during the year.
<i>Independent Variables</i>	
<i>High Exposure</i>	A binary variable that equals one if the firm shifts income out in any year from 2014 to 2016 and zero otherwise. The income shifting measure is calculated following prior literature (Klassen and Laplante 2012; De Simone et al. 2022).
<i>High Exposure_alt</i>	A binary variable that equals one if the firm shifts income out consistently for all the years from 2014 to 2016, and zero otherwise. The income shifting measure is calculated following prior literature (Klassen and Laplante 2012; De Simone et al. 2022).
<i>Post</i>	A binary variable that equals one for the years after 2017, and zero otherwise.
<i>ROA_EBITDA</i>	The ratio of EBITDA to the average total assets.
<i>Size</i>	The natural logarithm of the total assets.
<i>MTB</i>	The ratio of the market capitalization to the book value of equity.
<i>Sales Growth</i>	The growth rate of sales.
<i>Leverage</i>	The ratio of long-term debt to the lagged total assets.
<i>Inventory</i>	The ratio of inventory to the lagged total assets.
<i>PPE</i>	The ratio of property, plant, and equipment to the lagged total assets.
<i>CAPX</i>	The ratio of capital expenditure to the lagged total assets.

Appendix I (continued)

Variable	Definition
<i>Independent Variables (continued)</i>	
<i>Intang</i>	The ratio of intangible assets to the lagged total assets.
<i>RD</i>	The ratio of research and development expenditure to the lagged total assets.
<i>Foreign Income</i>	The ratio of pre-tax foreign income to the average total assets.
<i>Change NOL</i>	The annual change in the net operating loss carryforward.
<i>NOL</i>	A binary variable that equals one if the firm has a positive value for the net operating loss carryforward.
<i>Cross-sectional Variables</i>	
<i>Low Profitability</i>	A binary variable that equals one if the firm's <i>ROA_EBITDA</i> is in the bottom quartile of the sample and zero otherwise.
<i>Low R&D</i>	A binary variable that equals one if the firm's R&D intensity (<i>RD</i>) is in the bottom quartile of the sample and zero otherwise.
<i>Low Import Shipping Cost</i>	A binary variable that equals one if the firm's industry-year import shipping costs are below the top quartile of the sample and zero otherwise. The shipping costs are measured at the industry-year level following Barrot et al. (2019).
<i>Additional variables</i>	
<i>Emission</i>	The natural logarithm of the total emissions (in pounds) produced in all plants owned by the firm during the year.
<i>Job Posting</i>	The inverse hyperbolic sine of the total number of online job postings issued by the firm during the year.
<i>Job Posting, High Skill</i>	The inverse hyperbolic sine of the total number of high-skill online job postings issued by the firm during the year.
<i>Job Posting, Low Skill</i>	The inverse hyperbolic sine of the total number of low-skill online job postings issued by the firm during the year.
<i>VertInteg</i>	The vertical integration score of the firm from Frésard et al. (2020).
<i>IntersegSales</i>	The inverse hyperbolic sine of the total amount of intersegment "eliminations."
<i>IntersegSales (Bus)</i>	The inverse hyperbolic sine of the total amount of intersegment "eliminations," from business segments only.
<i>IntersegSales (Geo)</i>	The inverse hyperbolic sine of the total amount of intersegment "eliminations," from geographic segments only.
<i>Price (Kg)</i>	The inverse hyperbolic sine of the total price of export (import) scaled by the weight, shipped to (from) a given foreign jurisdiction from (to) the firm during the year.
<i>Price (Volume)</i>	The inverse hyperbolic sine of the total price of export (import) scaled by the volume, shipped to (from) a given foreign jurisdiction from (to) the firm during the year.

Figure 1 Dynamic Effects on Trade Volume

This figure presents the dynamic effect of U.S. tax reform on international trade. Panel A presents the dynamic effect on exports. To estimate the dynamic effects, we replace the *High Exposure* $\times$ *Post* indicator from Equation (1) with interaction terms between the *High Exposure* indicator and event year indicators from years $t - 3$ to $t + 3$ around event year t . We omit year $t - 1$, which serves as our benchmark. In this figure, we plot the estimated interaction coefficients and their associated 95% confidence intervals.

Panel A: Export Trade Dynamics

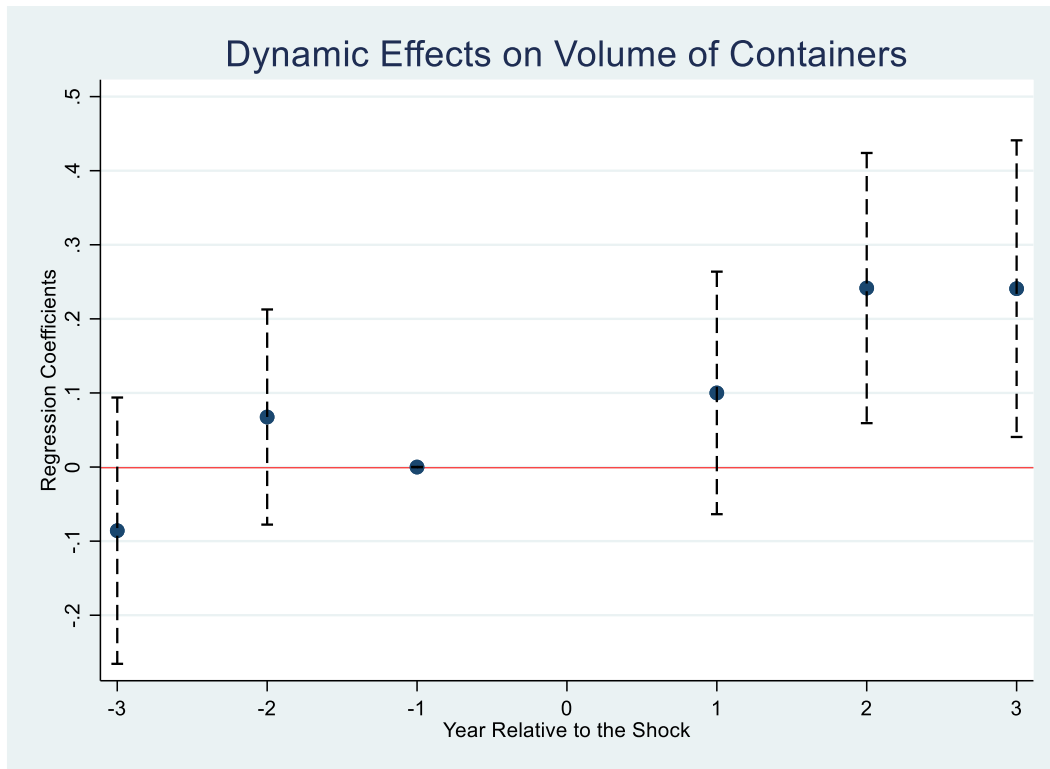


Figure 1 Dynamic Effects on Trade Volume (continued)

This figure presents the dynamic effect of U.S. tax reform on international trade. Panel B presents the dynamic effect on imports. To estimate the dynamic effects, we replace the *High Exposure* $\times$ *Post* indicator from Equation (1) with interaction terms between the *High Exposure* indicator and event year indicators from years $t - 3$ to $t + 3$ around event year t . We omit year $t - 1$, which serves as our benchmark. In this figure, we plot the estimated interaction coefficients and their associated 95% confidence intervals.

Panel B: Import Trade Dynamics

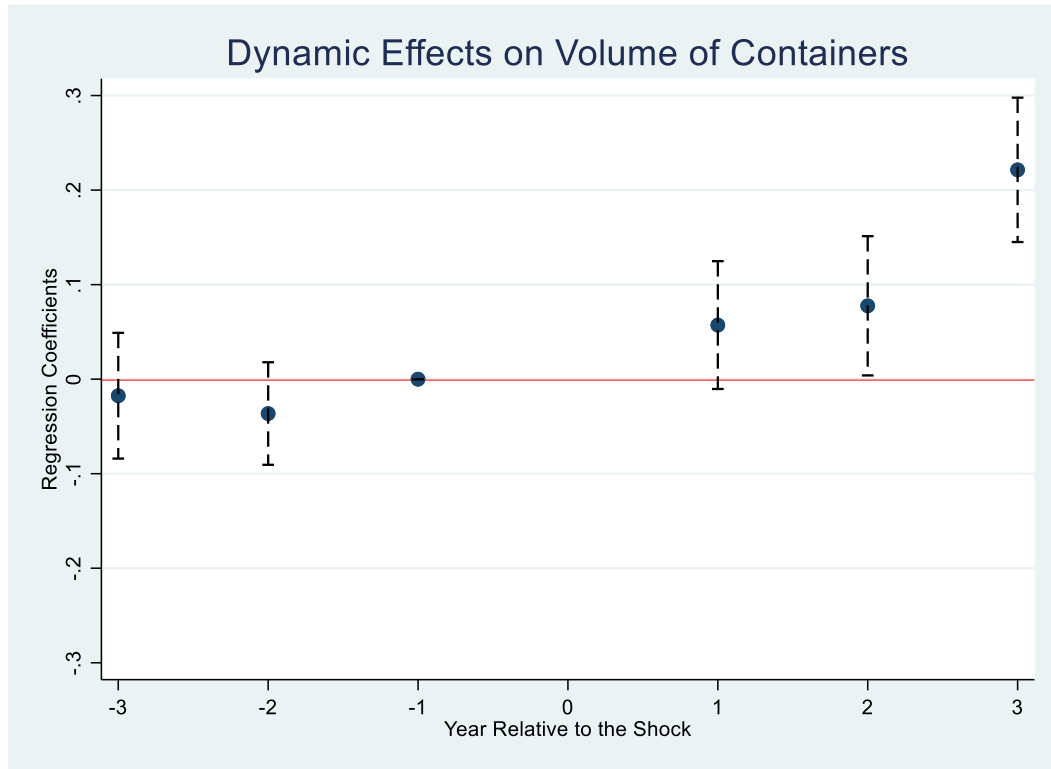


Table 1 Sample Selection and Distribution**Panel A: Sample Selection for the Export Sample**

Procedure	#Firms	#Observations
Compustat firms incorporated in the United States, firm-jurisdictions observations from 2014 to 2020	677	43,359
(–) Observations in 2017	0	6,164
(–) Trades to China or the United States	19	1,502
(–) Not matched with the income shifting measure	342	15,970
(–) Not matched with peers in the sample*	161	8,636
(–) Missing data requirements for the control variables in Compustat or singleton observations	7	333
Final sample	148	10,754

Panel B: Sample Selection for the Import Sample

Procedure	#Firms	#Observations
Compustat firms incorporated in the United States, firm-jurisdictions observations from 2014 to 2020	2,571	152,675
(–) Observations in 2017	0	21,833
(–) Trades from China or the United States	273	12,665
(–) Not matched with the income shifting measure	1,582	63,523
(–) Not matched with peers in the sample*	195	7,456
(–) Missing data requirements for the control variables in Compustat or singleton observations	21	1,634
Final sample	500	45,564

*To enhance the comparability of the treatment and control groups, we impose a matching requirement based on industry classification and foreign operations. For each firm in the treatment group (i.e., a firm that shifts income out in any year from 2014 to 2016), we require the inclusion of at least one control firm (i.e., a firm that did not shift income out in any year from 2014 to 2016) that belongs to the same industry classification (Fama-French 49 industries) and has similar foreign operation activities prior to the TCJA (i.e., within the same quartile of foreign income in the pre-period).

Panel C: Sample Distribution by Industry (Fama-French 49 industries)

Importer/Exporter Industry	Export Sample	Import Sample
Food Products	7.59	1.86
Beer & Liquor	1.00	0.40
Recreation	N/A	0.21
Entertainment	N/A	1.00
Printing and Publishing	N/A	0.36
Consumer Goods	4.20	3.20
Apparel	1.58	7.43
Medical Equipment	0.28	4.04
Pharmaceutical Products	1.73	2.67
Chemicals	24.80	8.32
Rubber and Plastic Products	0.81	0.22
Textiles	0.37	N/A
Construction Materials	3.43	2.76
Construction	N/A	0.61
Steel Works Etc	2.16	1.79
Fabricated Products	N/A	0.00
Machinery	12.96	13.83
Electrical Equipment	1.63	2.80
Automobiles and Trucks	10.61	6.70
Aircraft	0.55	0.00
Non-Metallic and Industrial Metal Mining	N/A	0.26
Petroleum and Natural Gas	2.44	5.64
Personal Services	N/A	0.17
Business Services	1.20	1.53
Computer Hardware	N/A	4.67
Computer Software	N/A	2.48
Electronic Equipment	0.61	6.95
Measuring and Control Equipment	1.28	2.92
Business Supplies	13.76	4.27
Transportation	0.52	0.69
Wholesale	5.50	5.24
Retail	N/A	4.83
Restaurants, Hotels, Motels	N/A	1.25
Banking	N/A	0.21
Insurance	N/A	0.09
Trading	N/A	0.19
Almost Nothing	0.98	0.38

Table 2 Summary Statistics

Panel A: Summary Statistics, Export Sample

Variables	Obs.	Mean	Std	p50	p75	p90	p95	p99
<i>Export Volume</i>	10,754	1.185	2.014	0.000	2.095	4.543	5.875	7.594
<i>High Exposure</i>	10,754	0.401	0.490	0.000	1.000	1.000	1.000	1.000
<i>Post</i>	10,754	0.476	0.499	0.000	1.000	1.000	1.000	1.000
<i>ROA_EBITDA</i>	10,754	0.152	0.069	0.140	0.174	0.265	0.287	0.374
<i>Size</i>	10,754	9.056	1.527	8.787	9.933	11.158	12.022	12.460
<i>MTB</i>	10,754	3.069	16.254	2.708	4.400	7.682	10.974	33.804
<i>Sales Growth</i>	10,754	0.021	0.109	0.021	0.064	0.136	0.198	0.358
<i>Leverage</i>	10,754	0.307	0.155	0.295	0.398	0.473	0.573	0.876
<i>Inventory</i>	10,754	0.139	0.090	0.114	0.165	0.274	0.340	0.417
<i>PPE</i>	10,754	0.266	0.145	0.230	0.365	0.473	0.537	0.696
<i>CAPX</i>	10,754	0.044	0.026	0.040	0.054	0.076	0.104	0.138
<i>Intang</i>	10,754	0.240	0.196	0.186	0.380	0.498	0.614	0.886
<i>RD</i>	10,754	0.021	0.021	0.017	0.027	0.052	0.066	0.114
<i>Foreign Income</i>	10,754	0.041	0.031	0.039	0.061	0.081	0.099	0.139
<i>Change NOL</i>	10,754	-0.004	0.054	0.000	0.002	0.019	0.050	0.141
<i>NOL</i>	10,754	0.679	0.467	1.000	1.000	1.000	1.000	1.000

Panel B: Summary Statistics, Import Sample

Variables	Obs.	Mean	Std	p50	p75	p90	p95	p99
<i>Import Volume</i>	45,564	0.853	1.651	0.000	0.881	3.611	4.942	6.468
<i>High Exposure</i>	45,564	0.406	0.491	0.000	1.000	1.000	1.000	1.000
<i>Post</i>	45,564	0.471	0.499	0.000	1.000	1.000	1.000	1.000
<i>ROA_EBITDA</i>	45,564	0.143	0.070	0.137	0.175	0.238	0.275	0.367
<i>Size</i>	45,564	8.561	1.645	8.423	9.634	10.854	11.568	12.378
<i>MTB</i>	45,564	3.868	7.586	2.707	4.389	8.211	12.940	51.023
<i>Sales Growth</i>	45,564	0.040	0.129	0.036	0.097	0.188	0.260	0.483
<i>Leverage</i>	45,564	0.243	0.177	0.219	0.333	0.462	0.553	0.913
<i>Inventory</i>	45,564	0.150	0.105	0.129	0.201	0.291	0.354	0.501
<i>PPE</i>	45,564	0.238	0.178	0.192	0.315	0.490	0.612	0.869
<i>CAPX</i>	45,564	0.043	0.033	0.033	0.056	0.089	0.116	0.163
<i>Intang</i>	45,564	0.244	0.213	0.194	0.360	0.534	0.648	1.058
<i>RD</i>	45,564	0.027	0.034	0.016	0.040	0.073	0.104	0.162
<i>Foreign Income</i>	45,564	0.043	0.040	0.036	0.062	0.095	0.121	0.182
<i>Change NOL</i>	45,564	0.001	0.035	0.000	0.003	0.022	0.053	0.161
<i>NOL</i>	45,564	0.683	0.465	1.000	1.000	1.000	1.000	1.000

Table 3 The Effect of U.S. Tax Reform on Trade Volume

This table shows the effect of U.S. tax reform on trade volume. Panel A presents the results for exports, and Panel B presents the results for imports. The dependent variable in Panel A is *Export Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped to a given foreign jurisdiction from the firm during the year. The dependent variable in Panel B is *Import Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year. *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. All columns control for firm-jurisdiction and year fixed effects. All the variables are defined in Appendix I. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered at the firm-jurisdiction level and appear in parentheses. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

Panel A: The Effect on Exports

	<i>Export Volume</i>	
	(1)	(2)
<i>High Exposure</i> × <i>Post</i>	0.258*** (0.077)	0.200*** (0.077)
<i>ROA_EBITDA</i>		-3.965*** (0.649)
<i>Size</i>		0.095 (0.129)
<i>MTB</i>		0.004*** (0.000)
<i>Sales Growth</i>		0.053 (0.190)
<i>Leverage</i>		-0.676*** (0.219)
<i>Inventory</i>		0.446 (0.695)
<i>PPE</i>		1.504*** (0.503)
<i>CAPX</i>		-2.375* (1.422)
<i>Intang</i>		-0.118 (0.278)
<i>RD</i>		20.189*** (5.082)
<i>Foreign Income</i>		1.827* (0.944)
<i>Change NOL</i>		0.699** (0.326)
<i>NOL</i>		0.056 (0.091)
Firm-Jurisdiction FE	Yes	Yes
Year FE	Yes	Yes
Obs.	10,754	10,754
Adj. R ²	0.587	0.598

Table 3 The Effect of U.S. Tax Reform on Trade Volume

This table shows the effect of U.S. tax reform on trade volume. Panel A presents the results for exports, and Panel B presents the results for imports. The dependent variable in Panel A is *Export Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped to a given foreign jurisdiction from the firm during the year. The dependent variable in Panel B is *Import Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year. *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. All columns control for firm-jurisdiction and year fixed effects. All the variables are defined in Appendix I. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered at the firm-jurisdiction level and appear in parentheses. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

Panel B: The Effect on Imports

	<i>Import Volume</i>	
	(1)	(2)
<i>High Exposure</i> × <i>Post</i>	0.137*** (0.029)	0.134*** (0.029)
<i>ROA_EBITDA</i>		-0.630*** (0.210)
<i>Size</i>		-0.005 (0.037)
<i>MTB</i>		0.007*** (0.001)
<i>Sales Growth</i>		-0.060 (0.051)
<i>Leverage</i>		-0.192** (0.080)
<i>Inventory</i>		1.555*** (0.213)
<i>PPE</i>		-0.105 (0.133)
<i>CAPX</i>		1.483*** (0.403)
<i>Intang</i>		0.211** (0.084)
<i>RD</i>		-2.975*** (0.847)
<i>Foreign Income</i>		-0.481 (0.300)
<i>Change NOL</i>		-0.434*** (0.145)
<i>NOL</i>		-0.035 (0.030)
Firm-Jurisdiction FE	Yes	Yes
Year FE	Yes	Yes
Obs.	45,564	45,564
Adj. R ²	0.598	0.600

Table 4 Cross-sectional Test: Pre-tax Profitability

This table shows the effect of U.S. tax reform on exports and imports, conditional on profitability. In column (1), the dependent variable is *Export Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped to a given foreign jurisdiction from the firm during the year. In column (2), the dependent variable is *Import Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year. *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. *Low Profitability* is a binary variable that equals one if the firm's *ROA_EBITDA* is in the bottom quartile of the sample and zero otherwise. All columns control for firm-jurisdiction and year fixed effects. All the variables are defined in Appendix I. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered at the firm-jurisdiction level and appear in parentheses. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

	<i>Export Volume</i>	<i>Import Volume</i>
	(1)	(2)
<i>High Exposure</i> × <i>Post</i> × <i>Low Profitability</i>	-0.568*** (0.170)	-0.182*** (0.065)
<i>High Exposure</i> × <i>Low Profitability</i>	0.219* (0.132)	-0.031 (0.047)
<i>Post</i> × <i>Low Profitability</i>	0.577*** (0.115)	0.118*** (0.038)
<i>High Exposure</i> × <i>Post</i>	0.333*** (0.085)	0.184*** (0.034)
<i>Low Profitability</i>	-0.007 (0.105)	-0.090*** (0.032)
Controls	Yes	Yes
Firm-Jurisdiction FE	Yes	Yes
Year FE	Yes	Yes
Obs.	10,754	45,564
Adj. R ²	0.600	0.600

Table 5 Cross-sectional Test: Intangible Investment

This table shows the effect of U.S. tax reform on exports and imports, conditional on R&D intensity as a proxy for intangible investment. In column (1), the dependent variable is *Export Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped to a given foreign jurisdiction from the firm during the year. In column (2), the dependent variable is *Import Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year. *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. *Low R&D* is a binary variable that equals one if the firm's R&D intensity (*RD*) is in the bottom quartile of the sample and zero otherwise. All columns control for firm-jurisdiction and year fixed effects. All the variables are defined in Appendix I. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered at the firm-jurisdiction level and appear in parentheses. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

	<i>Export Volume</i>	<i>Import Volume</i>
	(1)	(2)
<i>High Exposure</i> × <i>Post</i> × <i>Low R&D</i>	-0.370** (0.171)	-0.133** (0.065)
<i>High Exposure</i> × <i>Low R&D</i>	0.704*** (0.220)	0.255 (0.209)
<i>Post</i> × <i>Low R&D</i>	0.090 (0.105)	-0.122*** (0.039)
<i>High Exposure</i> × <i>Post</i>	0.289*** (0.092)	0.160*** (0.035)
<i>Low R&D</i>	0.018 (0.141)	-0.238 (0.152)
Controls	Yes	Yes
Firm-Jurisdiction FE	Yes	Yes
Year FE	Yes	Yes
Obs.	10,754	45,564
Adj. R ²	0.599	0.600

Table 6 Cross-sectional Test: Import Shipping Costs

This table shows the effect of U.S. tax reform on imports, conditional on the import shipping costs. The dependent variable is *Import Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year. *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. *Low Import Shipping Cost* is a binary variable that equals one if the firm's industry-year import shipping costs are below the top quartile of the sample and zero otherwise. The industry-year import shipping costs are measured using industry import data following Barrot et al. (2019). All columns control for firm-jurisdiction and year fixed effects. All the variables are defined in Appendix I. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered at the firm-jurisdiction level and appear in parentheses. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

	<i>Import Volume</i>
	(1)
<i>High Exposure</i> × <i>Post</i> × <i>Low Import Shipping Cost</i>	0.206** (0.088)
<i>High Exposure</i> × <i>Low Import Shipping Cost</i>	-0.390*** (0.090)
<i>Post</i> × <i>Low Import Shipping Cost</i>	-0.238*** (0.059)
<i>High Exposure</i> × <i>Post</i>	0.037 (0.077)
<i>Low Import Shipping Cost</i>	0.254*** (0.058)
Controls	Yes
Firm-Jurisdiction FE	Yes
Year FE	Yes
Obs.	21,814
Adj. R ²	0.627

Table 7 Change in Real Production and Multinational Network Reorganization

This table shows the effect of the U.S. tax reform on firms' real production activities and multinational organization. The sample observation is a firm-year for firms included in the export or the import sample. All columns control for firm and year fixed effects. All the variables are defined in Appendix I. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered at the firm level and appear in parentheses. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

Panel A: Domestic Production Intensity

This panel shows the effect of U.S. tax reform on domestic production intensity. In Column (1), the dependent variable is *Emission*, defined as the natural logarithm of the total emissions (in pounds) produced in all plants owned by the firm during the year. Emission data is extracted from the TRI database. In Column (2), the dependent variable is *Job Posting*, defined as the inverse hyperbolic sine of the total number of online job postings issued by the firm during the year. In Column (3), the dependent variable is *Job Posting, High Skill*, defined as the inverse hyperbolic sine of the total number of high-skill online job postings issued by the firm during the year. In Column (4), the dependent variable is *Job Posting, Low Skill*, defined as the inverse hyperbolic sine of the total number of low-skill online job postings issued by the firm during the year. Job posting data is extracted from the Lightcast database. High-skill job postings are occupations in job zone "4" or "5". *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise.

	<i>Emission</i>	<i>Job Posting</i>	<i>Job Posting, High Skill</i>	<i>Job Posting, Low Skill</i>
	(1)	(2)	(3)	(4)
<i>High Exposure</i> × <i>Post</i>	-0.029 (0.235)	-0.000 (0.100)	-0.016 (0.091)	0.010 (0.100)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs.	1,161	2,766	2,766	2,766
Adj. R ²	0.910	0.928	0.930	0.921

Table 7 Change in Real Production and Multinational Network Reorganization**Panel B: Vertical Integration and Intersegment Sales**

This panel shows the effect of U.S. tax reform on vertical integration. In Column (1), the dependent variable is *VertInteg*, defined as the vertical integration score of the firm from Frésard et al. (2020). In Column (2), the dependent variable is *IntersegSales*, defined as the inverse hyperbolic sine of the total amount of intersegment “eliminations.” In Column (3), the dependent variable is *IntersegSales (Bus)*, defined as the inverse hyperbolic sine of the total amount of intersegment “eliminations,” from business segments only. In Column (4), the dependent variable is *IntersegSales (Geo)*, defined as the inverse hyperbolic sine of the total amount of intersegment “eliminations,” from geographic segments only. *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise.

	<i>VertInteg</i>	<i>IntersegSales</i>	<i>IntersegSales (Bus)</i>	<i>IntersegSales (Geo)</i>
	(1)	(2)	(3)	(4)
<i>High Exposure</i> × <i>Post</i>	0.001 (0.000)	0.236** (0.109)	0.029 (0.093)	0.143** (0.073)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs.	2,766	2,766	2,766	2,766
Adj. R ²	0.880	0.871	0.860	0.822

Table 8 Product Upgrading

This table shows the effect of U.S. tax reform on the price of export and import goods. The sample observation is a firm-jurisdiction-year. Columns (1) and (2) present results for the export sample. Columns (3) and (4) present results for the import sample. In Columns (1) and (3), the dependent variable is *Price (Kg)*, defined as the inverse hyperbolic sine of the total price of export (import) scaled by the weight, shipped to (from) a given foreign jurisdiction from (to) the firm during the year. In Columns (2) and (4), the dependent variable is *Price (Volume)*, defined as the inverse hyperbolic sine of the total price of export (import) scaled by the volume, shipped to (from) a given foreign jurisdiction from (to) the firm during the year. *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. All columns control for firm-jurisdiction and year fixed effects.

	<i>Export Price (Kg)</i>	<i>Export Price (Volume)</i>	<i>Import Price (Kg)</i>	<i>Import Price (Volume)</i>
	(1)	(2)	(3)	(4)
<i>High Exposure</i> × <i>Post</i>	0.056 (0.082)	-0.060 (0.107)	-0.090 (0.063)	-0.104 (0.069)
Controls	Yes	Yes	Yes	Yes
Firm-Jurisdiction FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs.	2,085	2,056	6,268	6,070
Adj. R ²	0.648	0.550	0.552	0.417

Table 9 Robustness Tests

This table shows the robustness tests of the effect of U.S. tax reform on exports and imports. All continuous variables are winsorized at the 1% and 99% levels. Standard errors are clustered at the firm-jurisdiction level and appear in parentheses. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

Panel A: Alternative Definition for the Treatment Firms

This panel reports the results using an alternative proxy for TCJA exposure. *High Exposure_alt* is a binary variable that equals one if the firm shifts income out consistently for all the years from 2014 to 2016, and zero otherwise. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. The income shifting measure is calculated following prior literature (Klassen and Laplante 2012; De Simone et al. 2022). In column (1), the dependent variable is *Export Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped to a given foreign jurisdiction from the firm during the year. In column (2), the dependent variable is *Import Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year. All columns control for firm-jurisdiction and year fixed effects. All the variables are defined in Appendix I.

	<i>Export Volume</i>	<i>Import Volume</i>
	(1)	(2)
<i>High Exposure_alt</i> × <i>Post</i>	0.405*** (0.090)	0.216*** (0.035)
Controls	Yes	Yes
Firm-Jurisdiction FE	Yes	Yes
Year FE	Yes	Yes
Obs.	6,953	32,674
Adj. R ²	0.655	0.600

Panel B: Poisson Regression

This panel reports the results using the Poisson regression model. The dependent variable in column (1) is *Export Raw*, defined as the raw volume of the containers (measured in TEU) shipped to a given foreign jurisdiction from the firm during the year. The dependent variable in column (2) is *Import Raw*, defined as the raw volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year. *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. All columns control for firm-jurisdiction and year fixed effects. All the variables are defined in Appendix I.

	<i>Export Raw</i>	<i>Import Raw</i>
	(1)	(2)
<i>High Exposure</i> × <i>Post</i>	0.349** (0.145)	0.275*** (0.082)
Controls	Yes	Yes
Firm-Jurisdiction FE	Yes	Yes
Year FE	Yes	Yes
Obs.	8,046	33,665
Adj. R ²	0.858	0.811

Panel C: Alternative Measures of Trade Volume

This panel reports the results using alternative measures for export and import activities based on the weight or value of the containers. In column (1) (column (2)), the dependent variables are *Export Weight* (*Import Weight*), defined as the inverse hyperbolic sine of the weight of the containers (measured in Kilograms) shipped to a given foreign jurisdiction from the firm during the year (shipped from a given foreign jurisdiction to the firm during the year). In column (3) (column (4)), the dependent variables are *Export Value* (*Import Value*), defined as the inverse hyperbolic sine of the value of the containers (measured in USD) shipped to a given foreign jurisdiction from the firm during the year (shipped from a given foreign jurisdiction to the firm during the year). *High Exposure* is a binary variable that equals one if the firm shifts income out in any year from 2014 to 2016 and zero otherwise. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. All columns control for firm-jurisdiction and year fixed effects. All the variables are defined in Appendix I.

	<i>Export Weight</i>	<i>Import Weight</i>	<i>Export Value</i>	<i>Import Value</i>
	(1)	(2)	(3)	(4)
<i>High Exposure</i> × <i>Post</i>	0.876*** (0.253)	0.370*** (0.105)	0.707*** (0.237)	0.236** (0.093)
Controls	Yes	Yes	Yes	Yes
Firm-Jurisdiction FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs.	10,754	45,564	10,754	45,564
Adj. R ²	0.449	0.460	0.490	0.486

Panel D: Including Trades with China

This panel reports the results using an alternative sample by including exports to (imports from) China. In column (1), the dependent variable is *Export Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped to a given foreign jurisdiction from the firm during the year. In column (2), the dependent variable is *Import Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year. *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. All columns control for firm-jurisdiction and year fixed effects. All the variables are defined in Appendix I.

	<i>Export Volume</i>	<i>Import Volume</i>
	(1)	(2)
<i>High Exposure</i> × <i>Post</i>	0.161** (0.076)	0.131*** (0.029)
Controls	Yes	Yes
Firm-Jurisdiction FE	Yes	Yes
Year FE	Yes	Yes
Obs.	11,592	48,034
Adj. R ²	0.611	0.633

Panel E: Controlling for Bilateral Tax Information Exchange with the U.S.

This panel reports the results after controlling for TIEA adoptions during our sample period. In column (1), the dependent variable is *Export Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped to a given foreign jurisdiction from the firm during the year. In column (2), the dependent variable is *Import Volume*, defined as the inverse hyperbolic sine of the volume of the containers (measured in TEU) shipped from a given foreign jurisdiction to the firm during the year. *High Exposure* is a binary variable that equals one if the firm is identified as an income shifter during the 2014–2016 period. *Post* is a binary variable that equals one for the years after 2017, and zero otherwise. *TIEA* is an indicator for jurisdictions implementing a TIEA with the United States during the sample period, and *Post_TIEA* is an indicator for the post-TIEA period. The three treated jurisdictions are Argentina (signed December 23, 2016), Costa Rica (signed May 23, 2018), and Singapore (signed November 13, 2018). All columns control for firm-jurisdiction and year fixed effects. All the variables are defined in Appendix I.

	<i>Export Volume</i>	<i>Import Volume</i>
	(1)	(2)
<i>High Exposure</i> × <i>Post</i>	0.200*** (0.077)	0.134*** (0.029)
<i>TIEA</i> × <i>Post_TIEA</i>	-0.088 (0.155)	-0.149** (0.072)
Controls	Yes	Yes
Firm-Jurisdiction FE	Yes	Yes
Year FE	Yes	Yes
Obs.	10,754	45,564
Adj. R ²	0.598	0.600